🧠 What is Moravec's Corner Detector?

Moravec's Corner Detector is one of the **earliest algorithms** used to detect **corners** in an image — a corner being a point where **two edges meet**, or where there is a **large change in intensity in all directions**.

It helps identify **key points** in an image that are good for **feature matching**, **object tracking**, and **image registration**.

🧮 How does it work?

Step 1: Select a window

Take a small square **window W of size n × n** centered around a pixel p.

Step 2: Shift the window

Move the window slightly in various directions (usually to its **8 neighboring pixels**) and compare how much the **intensity values change**.

Step 3: Compute the squared intensity difference

For each shift, calculate the **sum of squared differences** (SSD) between the original window and the shifted window.

$$C(p) = \min_{q \in N(p)} \|W(p) - W(q)\|^2$$

Here:

- $C(p)$: the **cornerness** at pixel `p`.
- $W(p)$: the intensity values inside the window centered at `p`.
- $W(q)$: the intensity values inside the window when it's moved to neighbor `q`.
- $\|\cdot\|^2$: squared difference between pixel values.
- $N(p)$: the neighborhood of `p` (usually 8 directions).

We take the **minimum** SSD across all shifts — if all shifts result in **significant intensity change**, then we likely have a **corner**.

📊 What do the values mean?

Let's understand what $C(p)$ tells us:

Region Type	What Happens When Window Shifts?	$C(p)$ Value	Interpretation
Flat Region	Almost no change in intensity	Low	Not a corner
Edge	Changes only in one direction	Low	Not a corner
Corner	Changes in all directions	High	Likely a corner ✅

📜 After Calculating C(p) for all pixels

1. **Thresholding**: Only keep pixels where C(p)C(p)C(p) is above a certain threshold (these are potential corners).
2. **Non-Maximum Suppression (NMS)**: From nearby corner candidates, keep only the pixel with the highest cornerness in a small region — to avoid duplicate detections.

✅ Important Remarks

- **Moravec's detector is isotropic**:
  - That means it's **independent of direction**.
  - It treats all directions equally (unlike some other detectors which are direction-aware).
- It was **later improved by Harris Corner Detector**, which uses gradients and avoids the min function (Harris uses a more mathematically elegant formulation).